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Notes on Classical Electron Dynamics: Lorentz–Dirac Equation

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ABSTRACT

1. THE LORENTZ EQUATION AND THE RUNAWAY PROBLEM

[Conventions, stated once here and used throughout: Gaussian units for electrodynamics, SI for mechanics, $c = 1$. Metric signature and the definition of u^α, a^α to be fixed here. Note that the 2006 text declared Gaussian units but evaluated τ_0 without the factor $1/4\pi\varepsilon_0$; the corrected value is used here from the start.]

1.1. *Electrodynamics does not supply an equation of motion for the charge*

Classical electrodynamics rests on three pillars: the conservation laws, chief among them the conservation of charge; Maxwell's equations; and the Lorentz force law. Maxwell's equations fix the field produced by a given distribution of charge and current. The Lorentz force law fixes the force exerted on a charge by a given field. Between them the two appear to account for everything an electrodynamical problem could ask for, and they do not: neither supplies the equation of motion of the charge itself.

The gap is structural rather than an oversight. To follow the motion of a charge one needs the force acting on it, hence the total field at its location, and the total field includes the field that the charge itself produces. Maxwell's equations say how that self-field is generated but not how it acts back; the Lorentz force law says how a field acts on a charge but is silent on whether the charge's own field belongs among the fields that act. Electrodynamics is complete as an account of how charges make fields and of how fields push charges, and it is incomplete exactly where the two meet on the same particle. Closing that gap is the subject of everything below.

1.2. *Three historical routes*

[Open with the dichotomy that organizes the three attempts: the point-charge picture against the extended-charge picture. Under the first, the self-field diverges on the world line and with it the Lorentz force, so the force expression has no meaning where it is needed — one may reply that the Lorentz force was only ever meant for external fields, but then nothing at all determines the self-force, and that is the difficulty of the picture. Under the second, like charges repel, so a charge distribution flies apart, contrary to the existence of bound charges. Then: (i) Poincaré (1906): extended picture plus a binding stress–energy tensor (Poincaré stresses) introduced solely to hold the charge together — a force postulated to resolve one contradiction and nothing else, which is why it found few takers. (ii) Lorentz (1909): uniformly charged sphere. (iii) Dirac (1938): point picture with momentum–energy conservation. Close by stating the choice adopted here — the point picture — and why: technical uniformity, a simple model with no charge structure, and it generalizes.]

1.3. *The Lorentz model and its three defects*

[The 4/3 factor in the electromagnetic momentum; the lack of manifest covariance; and, most importantly, the self-acceleration.]

1.4. *The Abraham–Lorentz equation and its runaway solutions*

[Derive or quote the nonrelativistic equation and exhibit the exponentially growing free solution. This is the disease that the rest of the notes tracks.]

$$m\dot{\mathbf{v}} - \tfrac{2}{3}e^2\ddot{\mathbf{v}} + \cdots = \mathbf{F}, \tag{1}$$

[Define the characteristic time here and give the number, since every later section uses it:]

$$\tau_0 = \frac{2e^2}{3mc^3} = 6.27 \times 10^{-24}\ \mathrm{s} \quad \text{(electron)}. \tag{2}$$

1.5. *Criteria for a physically acceptable equation of motion*

[The three criteria stated in the introduction of the 2006 text, to be used as the scorecard in Secs. 3, 4 and 5: (1) the motion follows from an ordinary differential equation solved with initial/boundary data; (2) a steadily moving particle perturbed by a pulse returns to steady motion, rather than running away to c or oscillating between ±c; (3) energy conservation and causality hold.]

2. THE LORENTZ–DIRAC EQUATION: DERIVATION

[Terminology: throughout these notes "Lorentz–Dirac" (LD, also called Abraham–Lorentz–Dirac) refers to the classical equation of motion of a point charge, never to Dirac's relativistic wave equation.]

2.1. *The world-tube momentum–energy balance*

[Dirac's construction: surround the world line by a thin tube, integrate the conservation law over it, and let the radius shrink. Set up the geometry and the surface integrals.]

2.2. *The undetermined four-vector B^μ*

[The tube calculation leaves a four-vector B^μ undetermined, restricted only by $v \cdot B = 0$. Dirac's choice is one option among many; this freedom is what Sec. 5 exploits.]

2.3. *The equation*

$$ma^\alpha = F^\alpha_{\text{ext}} + \tfrac{2}{3}e^2\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta. \tag{3}$$

[Identify the two pieces of the reaction term: the Schott term and the radiated-momentum term, and note that the projector enforces $u \cdot a = 0$.]

2.4. *The other route: separating the singular part*

[State without proof here that the same equation follows from computing the Lorentz force with $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$ as the potential. The equivalence of the two routes is proved in Sec. 4.1, and that proof is what opens the door to Sec. 4.]

2.5. *Two reservations about the derivation*

[As recorded in the 2006 text: (i) the form of B^μ is arbitrary, fixed only by $v \cdot B = 0$; (ii) both derivations evaluate the tube surface integral but not the cap integrals.]

3. THE LORENTZ–DIRAC EQUATION: DISCUSSION AND SOLUTIONS

3.1. *One-dimensional reduction*

[Parametrize by the rapidity q via $\dot{t} = \cosh q$, $\dot{x} = \sinh q$. Equation (3) collapses to a single scalar equation:]

$$\dot{q} - \tfrac{2}{3}\ddot{q} = F^{01}. \tag{4}$$

3.2. *The free solution and the runaway*

$$q = C_1 e^{3\tau/2} + C_2. \tag{5}$$

[$C_1 \neq 0$ is the runaway; $C_1 = 0$ has to be imposed by hand. Note that no initial condition on position and velocity alone selects it — the equation is third order.]

3.3. *Dirac's asymptotic condition and the price paid*

[Pushing the condition out to $\tau \to \pm\infty$ eliminates the runaway but introduces pre-acceleration: the particle starts moving before the force arrives, on a timescale τ_0. This trades criterion (2) for criterion (3) of Sec. 1.5.]

3.4. *Energy balance and the Schott term; hyperbolic motion*

[Where the energy sits during acceleration, and the standing puzzle of uniformly accelerated motion, in which the reaction term vanishes although the particle radiates.]

3.5. *Scorecard, and the fork in the road*

[Against the three criteria of Sec. 1.5, LD fails all three in one form or another. Two ways to modify it then present themselves, and they are the subjects of the next two sections:

- sideways — *keep the structure of the reaction term and open up its coefficient (Sec. 4);*
- upward — *keep the coefficient and raise the order of the equation by enlarging B^μ (Sec. 5).*

This contrast is the spine of the notes.]

4. SPLITTING OFF THE SINGULAR PART: THE λ–κ FAMILY

4.1. *The equivalence theorem*

[The motion obtained from momentum–energy conservation is the same as the motion obtained by applying the Lorentz force law with $\frac{1}{2}(A_{\mathrm{ret}} - A_{\mathrm{adv}})$. Give the elementary proof from the 2006 text (Taylor expansion of the retarded potential, term-by-term cancellation), which is more direct than the routes via Zhang Zongsui or Poisson.]

4.2. *Does the singular part act?*

[Construct the stress–energy tensor of $A_S = \frac{1}{2}(A_{\text{ret}} + A_{\text{adv}})$ and integrate it. Eq. (1.2.17) of the 2006 text:]

$$\frac{dP^\alpha}{d\tau} = \frac{a^\alpha}{2r} + \frac{2}{3}\dot{a}^\alpha. \tag{6}$$

[The 2006 text read the surviving second term as showing that the singular part is not *inert, and used that as the licence to loosen the split. Flag: this is the load-bearing step of the whole section, and it needs to be checked — see the closing remark of Sec. 4.7.]*

4.3. *The two-parameter family*

$$A^\alpha = \lambda A^\alpha_{\text{ret}} + \kappa A^\alpha_{\text{adv}}. \tag{7}$$

[Two branches are then singled out by requiring that the retained part have a sensible field-theoretic status.]

4.4. *Branch (a): $\lambda + \kappa = 0$*

[The effective part satisfies the homogeneous Maxwell equations. The free-particle a–v phase portrait has a closed-form solution:]

$$a = \left[-\frac{3}{8\kappa}\ln\frac{1+v}{1-v} + C\right](1-v^2)^{3/2}. \tag{8}$$

[$\kappa = -1/2$ recovers LD. The resulting equation of motion is Eq. (1.5.1) of the 2006 text:]

$$ma^\alpha = F^\alpha_{\text{ext}} - \frac{4\kappa}{3}e^2\left(\dot{a}^\alpha - a^2 u^\alpha\right). \tag{9}$$

[Behaviour: $\kappa < 0$ gives a convex curve and still runs away; $\kappa > 0$ gives a concave curve that must cross the v axis, so a perturbed particle decelerates *back to uniform motion. The 2006 text called this "Aristotle motion," since force then fixes velocity rather than acceleration.]*

4.5. *Branch (b): $1 - \lambda = -\kappa$*

[Retarded and advanced parts enter the discarded singular term on equal footing. Solve numerically in κ. Result: $\kappa = -1$ is the watershed. For $\kappa > -1$ the runaway survives; for $\kappa < -1$ a critical velocity v_c appears — a particle starting below v_c stays below it, one starting above decelerates stably. Reproduce the table:]

κ	-1.5	-2	-3	-500	-5000
v_c	0.3473	0.2261	0.1341	6.67×10^{-3}	6.67×10^{-4}

4.6. *An unresolved point*

[Extending branch (b) beyond one dimension runs into $\mathbf{n}\cdot\mathbf{v}$ being undefined at the position of the charge itself. The 2006 text suggested the replacement $\mathbf{n}\cdot\mathbf{v} \to \frac{1}{3}v$ but did not carry it out.]

4.7. *What this branch amounts to*

[Eq. (9) is nothing but the LD equation with the radiation-reaction coefficient $2e^2/3$ replaced by a free parameter $-4\kappa e^2/3$. State this plainly — it is the one result in these notes that outlives its original motivation, whatever becomes of Eq. (6).]

5. THE ELIEZER-TYPE SERIES

5.1. *A note on the name*

[Placed first, not in a footnote. The equations of this section follow an ansatz for B^μ taken from Eliezer's 1947 review. They are not *the equation now standardly called the Eliezer, or Eliezer–Ford–O'Connell, equation, which is obtained by order reduction, is free of runaways, and is a live candidate in present-day experiments. Confusing the two inverts the conclusion of this section.]*

5.2. *Expanding B^μ*

$$B_\mu = P_0 v_\mu + P_1 \dot{v}_\mu + P_2 \ddot{v}_\mu + \cdots + P_i v_\mu^{(i)} + \cdots \tag{10}$$

[Differentiating $v^\mu v_\mu = -1$ repeatedly generates a chain of identities whose coefficients form Pascal's triangle; the general term is in Appendix A.4 of the 2006 text. Substituting gives the generalized equation of motion, Eq. (1.3.3) there. Setting all $P_i = 0$ returns LD, Eq. (3).]

5.3. *Keeping P_2: a third-order equation*

$$\dddot{q} - B\ddot{q} + A\dot{q} - \tfrac{1}{2}\dot{q}^3 = 0, \qquad A = \frac{3m}{2k}, \quad B = \frac{e^2}{k}. \tag{11}$$

[Scan the (A, B) plane numerically. Reported outcome: for A and B of either common sign the particle accelerates to c on its own; for $A, B > 0$ with $A \gg B$ the behaviour is worse still, the velocity swinging at high frequency between $+c$ and $-c$. Figures 1-3-1 through 1-3-4 of the 2006 text.]

5.4. *Keeping P_2 and P_4: a fifth-order equation*

[Eq. (1.3.11) of the 2006 text, scanned over 22 parameter triples (A, B, C). Three classes of behaviour: oscillation between $\pm c$; monotonic approach to c; and a third class, neither reaching c nor returning to uniform motion, in which the speed pulses periodically within a bounded range — "much like an oscillator, except that the oscillator itself keeps moving forward."]

5.5. *Scorecard*

[No parameter choice in either the third- or the fifth-order family satisfies the criteria of Sec. 1.5. Close with the question the 2006 text closed on — whether nature would really select a more complicated B^μ — and leave it open here.]